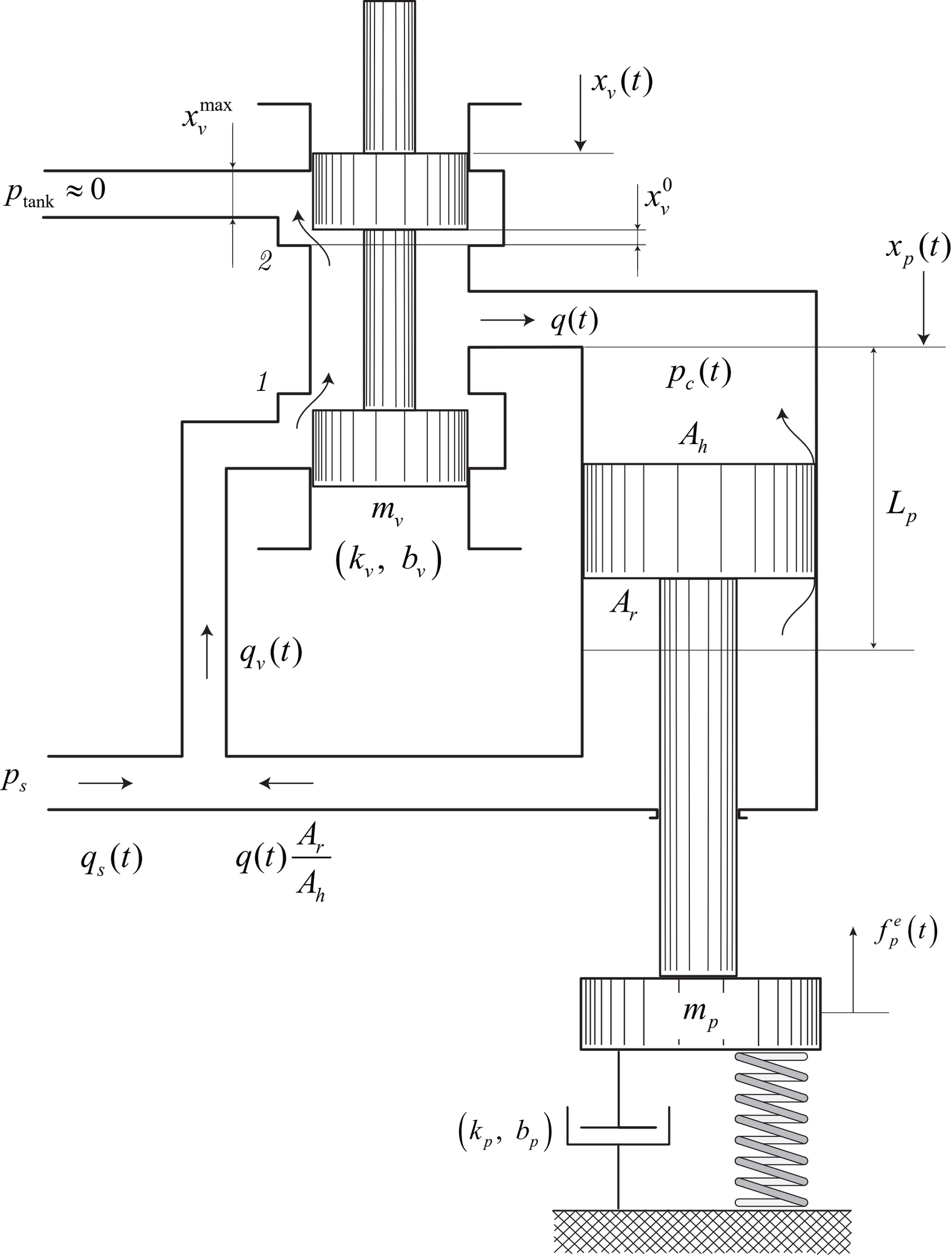

$x_v(t)$
$x_v^{\max}$
$p_{\text{tank}} \approx 0$
x_v^0
2
$x_p(t)$
$q(t)$
1
$p_c(t)$
A_h
m_v
$(k_v,\ b_v)$
L_p
A_r
$q_v(t)$
p_s
$q_s(t)$
$q(t)\dfrac{A_r}{A_h}$
$f_p^e(t)$
m_p
$(k_p,\ b_p)$